

SALESFORCE ORCHESTRATOR API DOCUMENTATION

Overview

This document describes the public API surface for the Salesforce orchestrator used in the hackathon implementation.

Package: salesforce_integration

Version: 1.0.1

Exports:

- salesforce_upsert
- salesforce_insert
- salesforce_update
- salesforce_delete
- SalesforceBulkWriteOperator
- SalesforceUpsertOperator

API Coverage Matrix

Capability	API/Primitive	Coverage Status	Notes
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Bulk upsert	salesforce_upsert	Covered	Supports external_id_field, optional wait
Bulk insert	salesforce_insert	Covered	Batch insert through Bulk API 2.0
Bulk update	salesforce_update	Covered	Expects Salesforce Id in records
Bulk delete	salesforce_delete	Covered	Expects Salesforce Id in records
Generic bulk orchestration	SalesforceBulkWriteOperator	Covered	insert/update/upsert/delete/hardDelete
Upsert operator wrapper	SalesforceUpsertOperator	Covered	Convenience wrapper over generic operator
UC connection auth via proxy	internal request path	Covered	Uses conn_id + UC proxy
Connection base-path fallback	internal request path	Covered	Retries alternate path style on 404
CSV LF line-ending compliance	internal serialization	Covered	Uses LF for Salesforce ingest compatibility
Task value propagation	functions/operators	Covered	Writes metadata best-effort to task values
Deferrable sensor for Salesforce	N/A in this repo	Not in scope	Intentionally removed for this implementation

Authentication Model

All API calls run through Databricks Unity Catalog Connection proxy endpoints.

Required runtime input:

- conn_id (for example: salesforce_m2m_conn)

No Salesforce credentials are embedded in source code.

Functions API

All functions are designed for Databricks Python Operator task parameter binding.
records is accepted as JSON string input and converted internally.

salesforce_upsert(object_name: str, external_id_field: str, records: str, conn_id: str,
api_version: str = "v60.0", wait_for_completion: bool = False) -> dict

Creates a Bulk API 2.0 upsert job, uploads records, and closes the job.

Returns:

- job_id
- object
- operation
- state
- records_total
- api_version
- conn_id
- plus records_processed/records_failed when wait_for_completion=True

salesforce_insert(object_name: str, records: str, conn_id: str, api_version: str = "v60.0",
wait_for_completion: bool = False) -> dict

Bulk insert flow for new records.

salesforce_update(object_name: str, records: str, conn_id: str, api_version: str = "v60.0",
wait_for_completion: bool = False) -> dict

Bulk update flow for existing records (records should include Salesforce Id).

salesforce_delete(object_name: str, records: str, conn_id: str, api_version: str = "v60.0",
wait_for_completion: bool = False) -> dict

Bulk delete flow for existing records (records should include Salesforce Id).

Function Runtime Behavior

Common internal flow for all write functions:

1. Coerce and validate records payload
2. Convert records to CSV with Salesforce-compatible LF line endings
3. Create Bulk API ingest job
4. Upload batch CSV
5. Set job state to UploadComplete
6. Optionally poll to terminal state when wait_for_completion=True

Robustness features:

- UC proxy path fallback for host-only vs base_path-style connections
- best-effort task value writes (salesforce_job_id, salesforce_object, salesforce_operation)
- explicit error raising on create/upload/close failures

Operators API

SalesforceBulkWriteOperator

Full lifecycle operator for external orchestration:

- open() -> creates and submits Salesforce Bulk job
- poll() -> checks job state and defers between checks
- close() -> attempts cleanup/abort for non-terminal jobs

Supported operations:

- insert
- update
- upsert
- delete
- hardDelete

Constructor parameters:

- object_name: str
- operation: str
- records: str
- conn_id: str
- task_key: str
- external_id_field: str = ""
- api_version: str = "v60.0"
- poll_interval_minutes: int = 1

Poll outcomes:

- JobComplete -> SensorResult.completed()
- Failed / Aborted -> raises exception
- transient states -> SensorResult.deferred(...)

SalesforceUpsertOperator

Convenience wrapper over SalesforceBulkWriteOperator with fixed operation="upsert".

Constructor:

- object_name
- external_id_field
- records
- conn_id
- task_key
- api_version
- poll_interval_minutes

Databricks Task Usage Examples

Function task example

python_operator_task:

main: salesforce_integration.salesforce_upsert

parameters:

- name: object_name
value: "Account"
- name: external_id_field
value: "Account_Number_External__c"
- name: records
value: '[{"Account_Number_External__c": "A001", "Name": "Acme"}]'
- name: conn_id
value: "salesforce_m2m_conn"
- name: wait_for_completion
value: "true"

Operator task example

python_operator_task:

main: salesforce_integration.SalesforceBulkWriteOperator

parameters:

- name: object_name
value: "Account"
- name: operation
value: "upsert"
- name: external_id_field
value: "Account_Number_External__c"
- name: records
value: '[{"Account_Number_External__c": "A001", "Name": "Acme"}]'
- name: conn_id
value: "salesforce_m2m_conn"
- name: task_key
value: "upsert_with_operator"

Error Model

Failures surface as Python exceptions with contextual messages, including:

- UC proxy request failures (status + response text)
- validation failures for payload/schema
- terminal Salesforce job failures (Failed, Aborted)

Deployment and Invocation

```
python3 setup.py bdist_wheel
databricks bundle validate --profile dogfood
databricks bundle deploy --profile dogfood
```

```
# Function workflow
databricks bundle run salesforce_full_workflow --profile dogfood
```

```
# Operator workflow
databricks bundle run salesforce_upsert_operator --profile dogfood
```

Notes

- This implementation is function/operator only (no Salesforce sensor in repo scope).
- Designed for hackathon demo plus production-oriented extension.